

PAU PYRENEES AP, France

WMO: 076100

Lat: 43.3850N

Lon: 0.4164W

Elev: 188

StdP: 99.09

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.7	-2.1	-6.0	2.3	-1.4	-4.4	2.7	-0.1	10.8	10.2	9.2	10.4	1.5	120	0.436

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.7	32.1	22.0	30.1	21.2	28.3	20.4	23.4	30.0	22.3	28.4	21.3	26.7	2.6	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.2	16.3	27.2	20.2	15.2	25.2	19.4	14.4	23.8	70.6	30.1	66.2	28.5	62.5	26.8	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.5	7.2	6.0	-6.1	35.8	1.9	1.7	-7.5	37.1	-8.6	38.1	-9.6	39.1	-11.0	40.4		
			-6.5	25.7	1.9	0.9	-7.9	26.3	-9.0	26.9	-10.0	27.4	-11.4	28.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.6	6.7	7.3	10.2	12.3	15.7	19.0	20.6	20.7	18.1	15.2	9.7	7.3
	DBStd	6.13	3.55	3.83	3.47	3.22	3.54	3.47	2.97	3.07	3.05	3.49	3.51	3.83
	HDD10.0	403	111	89	39	13	2	0	0	0	0	4	47	97
	HDD18.3	2009	361	310	252	183	97	34	12	11	42	108	261	341
	CDD10.0	1713	8	12	47	82	178	269	328	332	242	164	37	14
	CDD18.3	277	0	0	0	1	15	53	82	84	33	9	0	0
	CDH23.3	2462	0	1	5	34	177	506	658	679	305	95	2	0
	CDH26.7	802	0	0	0	4	44	180	222	254	82	16	0	0
Wind	WSAvg	2.4	2.5	2.6	2.8	2.8	2.7	2.6	2.4	2.2	2.0	2.0	2.3	2.3
Precipitation	PrecAvg	1153	107	93	88	115	111	87	73	70	82	96	135	96
	PrecMax	1625	252	199	209	234	254	279	137	185	152	248	250	173
	PrecMin	916	8	18	12	43	40	17	18	21	25	40	21	8
	PrecStd	174	63	57	50	50	44	50	33	37	38	46	60	46
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.7	21.2	24.0	26.9	30.7	34.0	34.0	35.5	31.6	28.8	22.3	19.2
		MCWB	12.4	12.2	13.6	16.8	19.5	22.6	23.3	22.6	20.7	18.3	14.9	12.6
	2%	DB	15.9	18.0	21.5	23.9	27.6	31.0	31.3	32.0	29.0	26.0	19.8	16.9
		MCWB	11.0	11.3	13.2	15.1	18.4	21.0	22.4	22.7	20.0	17.9	14.1	11.5
	5%	DB	13.8	15.8	19.2	21.4	25.1	28.6	29.1	29.3	27.0	23.7	17.1	14.9
		MCWB	10.2	10.4	12.4	14.2	17.5	20.4	21.4	21.7	19.3	17.0	12.7	10.7
	10%	DB	12.1	13.3	16.9	19.0	22.8	26.1	27.1	27.1	24.7	21.4	15.0	12.9
		MCWB	9.7	9.5	11.3	13.1	16.4	19.5	20.5	20.7	18.4	16.2	12.1	10.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.3	13.4	15.2	17.7	21.0	24.1	24.7	24.7	22.6	19.9	16.5	13.6
		MCDB	16.3	18.5	21.3	24.5	28.3	30.9	31.6	31.3	29.2	25.4	20.0	16.9
	2%	WB	12.0	12.1	13.8	16.0	19.4	22.3	23.3	23.4	21.1	18.6	14.7	12.4
		MCDB	14.4	16.3	19.5	22.2	25.9	29.1	29.8	29.9	27.0	23.8	18.1	15.3
	5%	WB	11.0	11.0	12.7	14.8	18.1	21.0	22.1	22.3	20.0	17.7	13.6	11.4
		MCDB	13.1	14.2	17.8	20.0	23.8	27.1	27.9	28.3	24.9	22.1	16.3	13.9
	10%	WB	9.9	10.0	11.8	13.6	16.9	19.8	21.0	21.1	19.0	16.8	12.4	10.5
		MCDB	11.7	12.9	16.1	17.9	21.6	25.3	25.9	26.0	23.3	20.4	14.6	12.6
Mean Daily Temperature Range	5% DB	MDBR	8.5	9.4	10.3	10.1	10.0	10.3	10.3	10.7	11.1	10.6	8.8	9.0
		MCDBR	11.6	14.2	14.7	14.8	14.6	15.3	14.9	15.1	14.8	14.2	12.6	12.0
		MCWBR	7.5	8.4	7.7	7.5	6.8	6.8	7.0	6.8	6.7	7.1	7.5	7.5
	5% WB	MCDBR	9.6	11.4	12.6	13.1	12.9	13.1	13.1	13.3	12.5	11.9	10.4	9.9
		MCWBR	6.9	7.4	6.9	7.0	6.4	6.5	6.7	6.5	6.2	6.4	6.8	6.7
Clear-Sky Solar Irradiance	taub	0.310	0.325	0.361	0.386	0.397	0.409	0.400	0.400	0.373	0.352	0.326	0.307	
	taud	2.499	2.441	2.359	2.296	2.310	2.309	2.337	2.347	2.412	2.475	2.498	2.505	
	Ebn at Noon	827	871	877	879	876	863	867	855	854	829	799	795	
	Edh at Noon	71	88	108	125	127	127	122	117	102	85	71	65	
All-Sky Solar Radiation	RadAvg	1.52	2.40	3.59	4.51	5.24	5.87	5.91	5.26	4.30	2.85	1.76	1.40	
	RadStd	0.14	0.29	0.39	0.48	0.44	0.40	0.39	0.35	0.21	0.24	0.20	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page